Xianda Du

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Education

University of Waterloo — CGPA until August 2025: 4.0 out of 4.0 Waterloo, ON
Bachelor of Applied Science in Computer Engineering, Option in Artificial Intelligence *September 2022 – May 2027*

- Undergraduate Research Assistant on FGSM and Gaussian Noise Analysis for ResNet-18 with Prof. En-Hui Yang.
- Undergraduate Research Assistant with Prof. Wenhu Chen on **VIEScore2**, fine-tuning **vision-language models** to evaluate AI-generated image quality with **interpretable visual grounding** that jointly produces a score, a textual rationale, and a spatial mask of where the image fails.

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, Bash
Frameworks: FastAPI, Pydantic, LangChain, React, Vue, Express, Spring Boot, Tailwind
Developer Tools: GCP, AWS, Docker, Kubernetes, Git, REST/GraphQL APIs, Airflow, Scrapy
Machine Learning: PyTorch, TensorFlow, Transformers, RAG, vLLM, Scikit-learn, XGBoost, LightGBM, NumPy, Pandas, OpenAI Structured Outputs
Database: Supabase, MySQL, MongoDB, Redis, Snowflake, Qdrant, FAISS

Professional Experience

Machine Learning Engineer January 2026 – April 2026
StackAdapt *Toronto, ON*

- Shipped StackAdapt's **Precision Audience** LLM pipeline, automating audience generation from advertiser briefs across **9 data sources** × **10 geos** and replacing a manual, error-prone workflow with an end-to-end system using **GPT-5 structured outputs**, **Pydantic** schemas, and **async** orchestration with **Redis** job tracking.
- Eliminated silent data corruption by building a **4-layer hallucination detection** system (schema enforcement, **reference-grounding** against candidate sets, strict enum validation, and post-LLM **safeguard rules**), ensuring every audience segment in the output corresponds to a validated upstream source.
- Reduced unsafe ad targeting and improved reliability by designing **prompt hardening** (injection detection, 10-category sensitive-data taxonomy) and a typed **exception hierarchy** that maps failures to correct **HTTP semantics**, enabling per-geo **partial-success (207)** so multi-country requests degrade gracefully.
- Scaled audience coverage to **~12.7K segments** across multiple countries by building **hybrid vector retrieval** on **Qdrant** (**dense + BM25 + RRF**, geo-indexed prefilter) and an **end-to-end data pipeline** with ISO validation, diff-based incremental ingestion, and **snapshot/restore** rollback to S3.

Research Assistant (NLP Benchmarks) Oct 2024 – Present
McGill University - RBC Borealis AI *Montreal, QC - Toronto, ON*

- Co-first authored **WXImpactBench** (**ACL 2025**), the **first benchmark** probing whether frontier LLMs can read historical disaster narratives and recognize real-world societal harm across infrastructural, financial, ecological, agricultural, public-health, and political dimensions; distilled **53K+ OCR-scanned** newspaper articles down to **1.7K expert-labeled** samples via a **GPT-4o** post-OCR correction + LDA topic-modeling pipeline.
- Designed a dual-task evaluation suite covering **multi-label impact classification** and **ranking-based QA**, powered by a **GPT-4o pseudo-question generator** and **sliding-window re-ranker**; benchmarked **12 production LLMs** (GPT, DeepSeek, Llama, Mistral, Qwen, Gemma) on F1, row-wise accuracy, Recall@1, and nDCG@5, establishing a comparative leaderboard for LLM reasoning on domain-specific historical narratives.
- Co-first authored **WeatherArchive-Bench** (**SIGIR 2026**), a benchmark that **jointly evaluates retrieval and downstream RAG quality** on historical weather archives; assembled a **1M+ passage** corpus with 335 expert-validated queries and vulnerability/resilience labels co-developed with climate scientists under the **IPCC framework**.
- Conducted a head-to-head comparison of **13+ retrieval systems** spanning classical (BM25), dense (SBERT, SPLADE, ANCE, UniCoil), and **frontier embedding APIs** (OpenAI, Gemini, Qwen, Arctic, Granite) with cross-encoder reranking; Gemini Embedding 001 led at **95.8% Recall@100**, providing empirical guidance for retriever selection in domain-specific RAG.
- Evaluated **17 frontier LLMs** (GPT-4.1, Claude, Gemini, Llama-3, Qwen-2.5/3, Mistral, DeepSeek-V3) on RAG-based QA and 6-dimension vulnerability classification, running QA under both **retrieved** and **gold-passage** conditions to **quantify retrieval's contribution** to downstream answer quality via BLEU, ROUGE-L, and **BERTScore**.

Platform Developer September 2024 – December 2024
University of Waterloo, ECE Department *Waterloo, ON*

- Fine-tuned **Hugging Face** sentiment models using **TensorFlow/PyTorch** to **80%** accuracy; built a classification pipeline converting **PostgreSQL** embeddings on **GKE** through a custom **neural network** to categorize scraped media articles.

Projects

WatAI RAG Project Technical Lead April 2025 – December 2025

- Built a flexible **LangChain** layer that can switch between **Qwen, OpenAI, and Gemini**; supports **streaming** responses, **custom prompts**, per-user **API keys**, and simple guardrails (rate and token limits).
- Delivered a production **RAG** flow: ingest & chunk course PDFs/slides, embed and store in a **Supabase**, retrieve top passages (optional **hybrid** dense+BM25), **re-rank**, and answer with grounded **citations**.
- Added **agent** tools (retrieval, file parsing, syllabus/FAQ lookup) and a router that picks the best model by **speed, cost, and capability**; included timeouts, retries, and safe **fallbacks**.
- Set up **evaluation & monitoring**: retrieval metrics (**Recall@k, nDCG**), **LLM-as-judge** faithfulness checks, and run tracing (model, prompt version, retriever settings) to tune RAG with real chats.

Web 3.0 Quantitative Trading Challenge Jun 2025 – Nov 2025

- Framed 15-min crypto returns as **ranking** and **Weighted Spearman**; trained **TabNet**, **XGBoost**, **MLP**, and **LightGBM** via **Optuna**; finalized **Optuna LightGBM v10**, scoring **0.57 Weighted Spearman** on the public leaderboard.
- Implemented **leak-safe, rolling-window** features (RSI divergence, distance-to-high/low, volatility) via vectorized **NumPy/pandas**; importance screening kept **80** features, improving generalization/training speed.
- Built **8-model LightGBM ensemble** (varying depth/regularization/seeds/losses) with **weighted blending**; applied adaptive clipping/winsorization, normalization, smoothing to stabilize ranks and beat single models.